

## MINOR PROJECT – 2

### Customer Segmentation using K-Means Clustering (Unsupervised Machine Learning)

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#### 1. Problem Statement

Customer segmentation is one of the most important applications of data analytics in the retail industry. Shopping malls and retail businesses serve customers with different purchasing behaviors, income levels, and age groups. Treating all customers in the same way often leads to ineffective marketing strategies and lower customer satisfaction.

The objective of this project is to identify distinct groups of customers based on their Age, Annual Income, and Spending Score using an unsupervised machine learning approach. Since the dataset does not contain predefined labels, K-Means Clustering is used to discover natural customer segments. The identified customer groups help businesses understand customer behavior, design personalized marketing campaigns, improve customer satisfaction, and increase overall business performance.

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#### 2. Project Objective

The primary objective of this project is to perform customer segmentation using the K-Means Clustering algorithm. The project aims to identify meaningful customer groups by analyzing customer characteristics such as Age, Annual Income, and Spending Score. These customer segments help businesses understand customer behavior, improve marketing strategies, optimize promotional campaigns, and enhance customer relationship management.

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#### 3. Dataset Description

The dataset used in this project is the **Mall Customers Dataset**, which contains demographic information and spending behavior of customers visiting a shopping mall.

##### Dataset Details

- Number of Records: **200**
- Number of Features: **5**

##### Dataset Features

- **CustomerID** – Unique identifier for each customer.

- **Gender** – Gender of the customer (Male/Female).
- **Age** – Age of the customer.
- **Annual Income (k\$)** – Annual income of the customer in thousand dollars.
- **Spending Score (1–100)** – Score assigned by the mall based on customer purchasing behavior.

The dataset is suitable for clustering because it does not contain any predefined target labels, making it an ideal dataset for unsupervised machine learning.

#### Dataset Source

<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial>

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#### 4. Data Preprocessing

Before training the clustering model, several preprocessing steps were performed to improve data quality and ensure reliable clustering results.

The following preprocessing steps were applied:

- Loaded the dataset using the Pandas library.
- Displayed dataset information and summary statistics.
- Checked for missing values.
- Checked for duplicate records and removed duplicates if present.
- Selected important numerical features for clustering.
- Applied feature scaling using **StandardScaler** to normalize the numerical features.
- Prepared the dataset for clustering.

These preprocessing steps ensure that all selected features contribute equally during distance calculations in the K-Means algorithm.

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#### 5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics and distribution of the dataset before building the clustering model.

The following visualizations were created:

- Histogram of Age
- Histogram of Annual Income
- Histogram of Spending Score
- Gender Distribution Pie Chart
- Correlation Heatmap

- Pairplot between Age, Annual Income, and Spending Score

The analysis showed noticeable differences in customer income and spending behavior, indicating the presence of naturally occurring customer groups within the dataset.

**(Insert screenshots of all graphs here.)**

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## 6. Model Used

This project uses the **K-Means Clustering Algorithm**, which is one of the most widely used unsupervised machine learning algorithms for customer segmentation.

K-Means partitions the dataset into K different clusters by minimizing the distance between each data point and the corresponding cluster centroid.

The optimal number of clusters was determined using the **Elbow Method** and further validated using the **Silhouette Score**.

### Why K-Means?

- Simple to understand and implement.
  - Efficient for numerical datasets.
  - Produces meaningful customer segments.
  - Suitable for market and customer segmentation.
  - Computationally fast for medium-sized datasets.
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## 7. Training Process

The following steps were followed during model training:

1. Imported the required Python libraries.
2. Loaded the Mall Customers dataset.
3. Performed data preprocessing.
4. Selected Age, Annual Income, and Spending Score as clustering features.
5. Applied feature scaling using StandardScaler.
6. Used the Elbow Method to estimate the optimal number of clusters.
7. Calculated the Silhouette Score for cluster validation.
8. Trained the K-Means clustering model.
9. Assigned cluster labels to every customer.
10. Visualized the resulting customer clusters.
11. Generated cluster summaries and interpreted customer behavior.

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## 8. Evaluation Metrics and Results

Since customer segmentation is an **unsupervised learning problem**, traditional classification metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix cannot be used because the dataset does not contain actual class labels.

Instead, the clustering model was evaluated using the following techniques.

### Elbow Method

The Elbow Method was used to determine the optimal number of clusters by analyzing the Within-Cluster Sum of Squares (WCSS). The graph showed an elbow around **K = 5**, indicating that five clusters provide a good balance between model complexity and clustering quality.

### Silhouette Score

The Silhouette Score measures how well each data point fits within its assigned cluster compared to other clusters. A higher Silhouette Score indicates better-separated and more compact clusters.

### Inertia

Inertia measures the compactness of the clusters by calculating the total squared distance between data points and their assigned centroids. Lower inertia values generally indicate better clustering performance.

The clustering model successfully identified meaningful customer groups with distinct income levels and spending behavior.

**(Insert screenshots of Elbow Method, Silhouette Score graph, and Cluster Visualization here.)**

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## 9. Cluster Interpretation

The clustering results revealed different categories of customers based on their purchasing behavior.

- One cluster represents customers with **high income and high spending**, who are considered premium customers.
- Another cluster consists of customers with **high income but low spending**, indicating potential customers who may require targeted marketing.
- Some clusters represent **average-income customers with moderate spending behavior**.
- Another cluster contains **low-income customers with relatively lower spending scores**.

These customer segments can help businesses improve personalized marketing strategies, customer retention, product recommendations, and promotional campaigns.

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## 10. Conclusion

This project successfully implemented the K-Means Clustering algorithm to perform customer segmentation using an unsupervised machine learning approach.

The complete machine learning workflow, including data preprocessing, exploratory data analysis, feature scaling, model training, cluster visualization, and evaluation, was successfully completed.

The generated customer segments provide valuable business insights that can be used for targeted marketing, customer relationship management, personalized promotions, and business decision-making. Overall, the project demonstrates the practical application of unsupervised machine learning in solving real-world business problems.